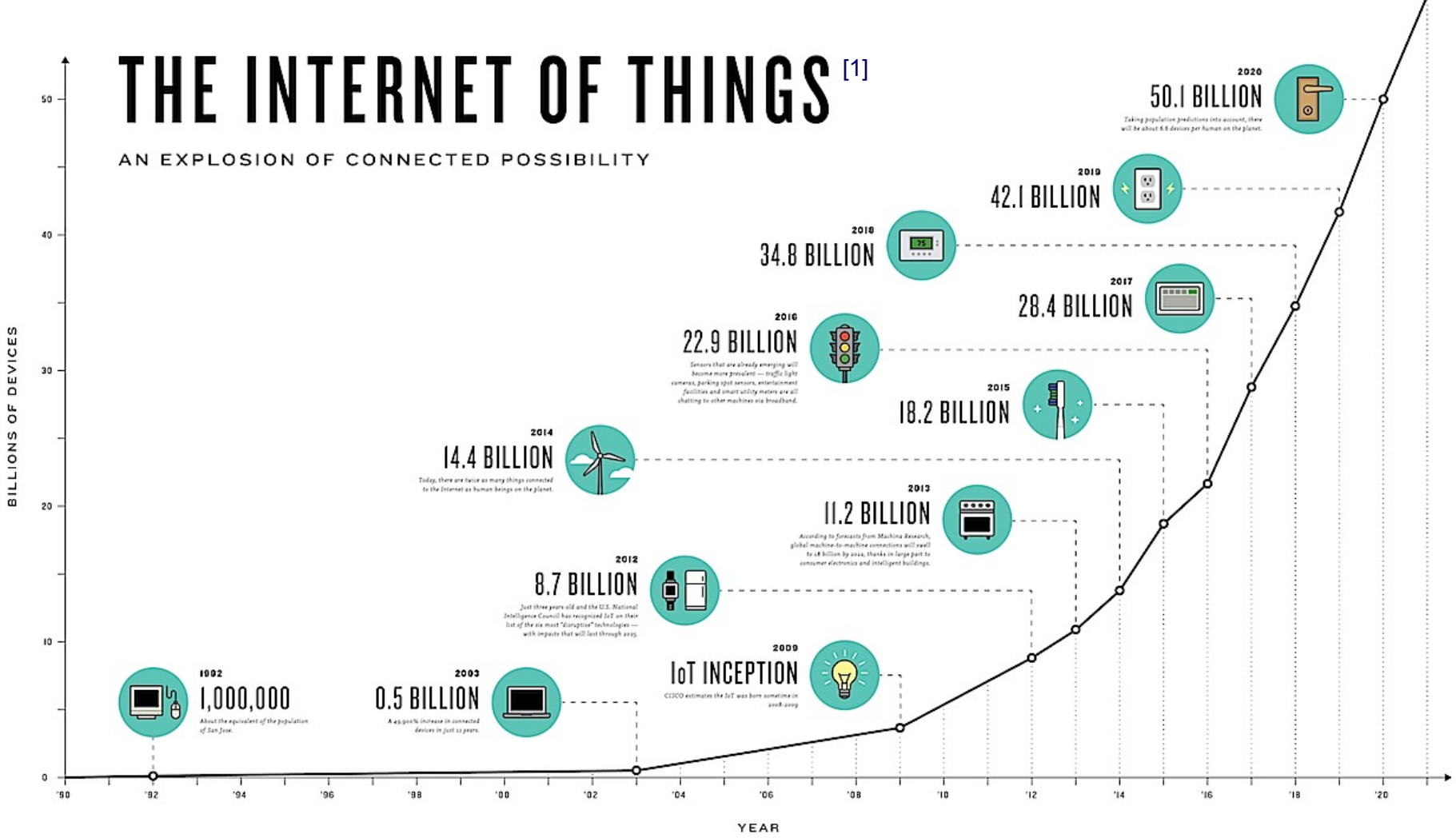
ZigBee Protocol Fuzzing

Internet of Things

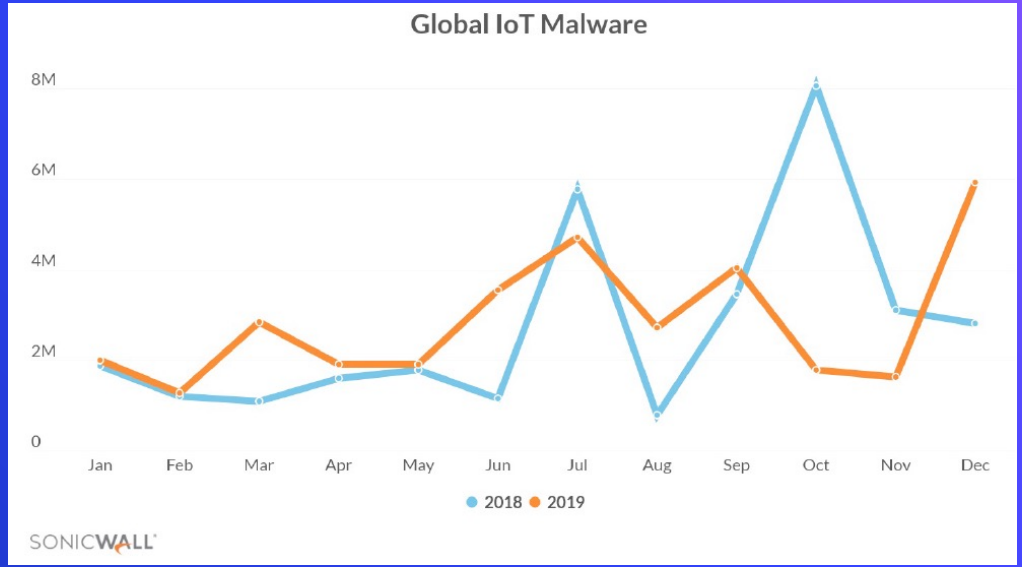
“ The Internet of things (IoT) is a system of interrelated computing devices, mechanical and digital machines, objects, animals or people that are provided with unique identifiers (UIDs) and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction. ”

- Wikipedia


THE INTERNET OF THINGS [1]
AN EXPLOSION OF CONNECTED POSSIBILITY
1992
1,000,000
About the equivalent of the population of San Jose.
2003
0.5 BILLION
A 49,900% increase in connected devices in just 10 years.
2009
IoT INCEPTION
CISCO estimates the IoT was born sometime in 2008-2009
2012
8.7 BILLION
Just three years old and the U.S. National Intelligence Council has recognized IoT on their list of the six most "disruptive" technologies — with impacts that will last through 2025.
2014
14.4 BILLION
Today, there are twice as many things connected to the Internet as human beings on the planet.
2013
11.2 BILLION
According to forecasts from Machina Research, global machine-to-machine connections will swell to 18 billion by 2022, thanks in large part to consumer electronics and intelligent buildings.
2016
22.9 BILLION
Sensors that are already emerging will become more prevalent — traffic light cameras, parking spot sensors, entertainment facilities and smart utility meters are all chatting to other machines via broadband.
2015
18.2 BILLION
2018
34.8 BILLION
2017
28.4 BILLION
2019
42.1 BILLION
2020
50.1 BILLION
Taking population predictions into account, there will be about 6.6 devices per human on the planet.
BILLIONS OF DEVICES
0
10
20
30
40
50
YEAR
'90
'92
'94
'96
'98
'00
'02
'04
'06
'08
'10
'12
'14
'16
'18
'20

IoT Attacks

The IoT is a treasure trove for cybercriminals.



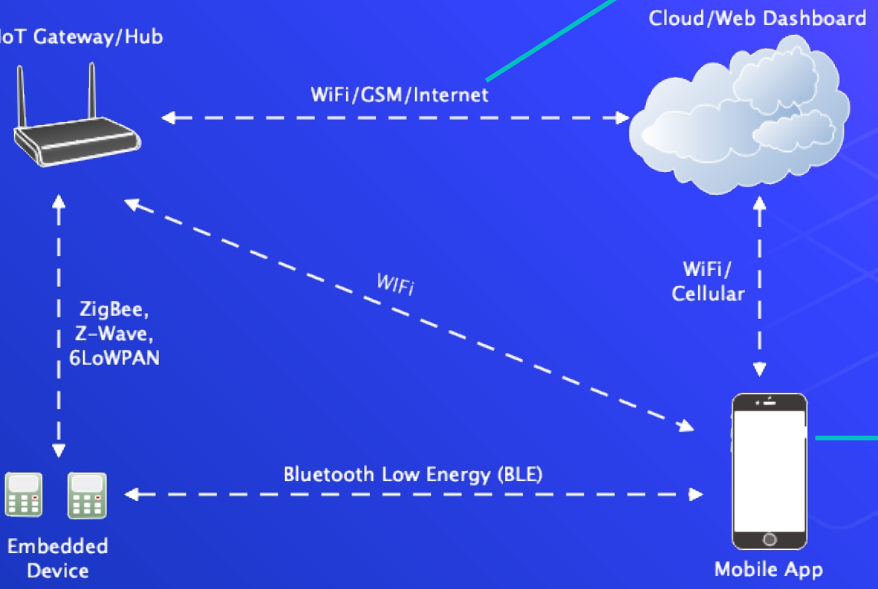
[2]

“Is Internet of Things Secure Enough?”

IoT Attack Surface [3]

Network Communication

Attacker could exploit unsecure network to manipulate transmitted message to control remote devices and return fake message to user.



Embedded Device

Attacker could manipulate the firmware to execute malicious codes, send user information to remote server.

Software & Application

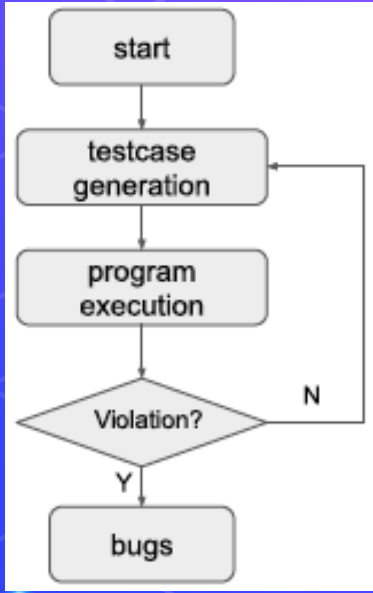
Attacker could hack application to pretend as an authorized user to remotely control the IoT devices.

ZigBee Protocol

- ZigBee is a wireless personal area network (PAN) standard that is designed for low-power, low-resource embedded devices.

ZigBee Protocol Fuzzing

- Fuzzing – randomly generate test data to target to find vulnerabilities.
- No physical devices – we present a pure software environment to simulator various IoT devices.
- We propose a testing framework for ZigBee protocol implementation to find its security issues.
- We add heuristic information to guide mutation of current protocol fuzzer for high effectiveness.



Thanks!

Any questions?

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Poster session: ZigBee Protocol Fuzzing

Reference

1. https://opentechdiary.wordpress.com/category/iot-basics/, accessed by Feb 2020
2. https://www.sonicwall.com/resources/2020-cyber-threat-report-pdf/, accessed by Feb 2020
3. https://resources.infosecinstitute.com/getting-started-with-iot-security-mapping-the-attack-surface/#gref